

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Student Assessment Document

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10: SUB-STRAND 1.6 PLANT GROWTH AND DEVELOPMENT: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

In this Final Explanation you will use everything you have learned in Lessons 1 to 8 to answer the driving question: Why do seeds from the same garden grow and develop so differently, and what controls how a plant grows from a tiny seed into a tall tree or a flowering herb? Remember the Kisumu farmer who planted maize and beans in the same bed, on the same day, in the same soil, with equal watering. Answer each section in the blank space under the prompt. Write in full sentences, use the correct science words (for example dormancy, epigeal, hypogeal, apical meristem, vascular cambium, auxin, abscisic acid), and refer to the evidence you saw in class. You may draw and label small diagrams where a prompt asks for one. Each answer should link back to the Kisumu farmer's garden.

Section 1: Why Did Some Stored Seeds Refuse to Germinate?

The Kisumu farmer found that some seeds from a packet stored for two seasons did not germinate, even when given water and warmth.

(a) Explain what dormancy is and say whether a dormant seed is dead or alive. (b) Describe TWO causes of dormancy that could explain why the stored seeds failed to germinate even though they had water and warmth. (c) Describe TWO ways the farmer could break dormancy before planting, linking each to a practice used by farmers in Kisumu. (d) Name the three external conditions a seed needs for germination, and state what each one does inside the seed.

Write your answer here:

Section 2: Why Do Maize and Bean Seedlings Emerge Differently?

Within a week the Kisumu farmer saw bean seedlings push their seed leaves above the soil in an arch, while maize sent up a single green spike and the seed stayed underground.

(a) Name the type of germination shown by beans and the type shown by maize. (b) Explain, using the words hypocotyl, epicotyl, cotyledon and plumule, how each type of seedling gets above the soil. (c) Explain one advantage of each strategy for the young seedling. (d) The bean and maize seeds were planted in the same soil with the same water. What does this tell you about the cause of their different emergence, and which part of the seedling's structure differs between the two?

Write your answer here:

Section 3: Getting Taller and Getting Thicker

Next to the bean plants in the farmer's garden are mango and avocado trees planted years earlier. They have thick, woody trunks, while the bean plants stay thin.

(a) Explain how primary growth and secondary growth are different, naming the tissue responsible for each. (b) Describe what the vascular cambium produces each growing season and how this forms the rings seen in the mango trunk photograph. (c) Explain why bean plants, sukuma wiki and maize do not develop thick woody trunks. (d) What can a person find out by counting the rings in a mango trunk?

Write your answer here:

Section 4: Chemical Control and the Environment

(a) Match each hormone to ONE job it does in the plant: auxins, gibberellins, cytokinins, abscisic acid, ethylene. **(b)** Sukuma wiki grown near a window bends toward the light. Explain how auxin causes this. **(c)** In a Kisumu market, mangoes stored in a crate on their own ripen slowly, but mangoes stored in a crate with ripe bananas ripen much faster. Explain why. **(d)** Choose ONE of the regional case studies (tea in Kericho, maize in the Rift Valley, or mangroves in Mombasa). Explain how water, temperature, light or nutrients affect how plants grow there, and how these factors work together with hormones or growth type.

Write your answer here:

Section 5: Explaining a New Problem and the Whole Phenomenon

A farmer in Nakuru plants bean seeds in a shaded corner of her plot. After two weeks the seedlings are much taller than seedlings in the sunny part of the plot, but their stems are thin and weak, their leaves are a washed-out colour, and they have few side branches.

(a) Name the process that has caused these differences and the environmental factor involved. (b) Use auxin, gibberellin and apical dominance to explain why the shaded seedlings are so long and thin with little branching. (c) Suggest TWO actions the farmer could take and explain how each works. (d) Now answer the driving question in 4 to 6 sentences: Why do seeds from the same garden grow and develop so differently, and what controls how a plant grows from a tiny seed into a tall tree or a flowering herb? Use ideas from dormancy, germination type, growth type, hormones and the environment.

Write your answer here:

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of science ideas (dormancy, germination, growth, hormones)	Uses all key ideas accurately and in detail, including correct terms such as epigeal, hypogeal, apical meristem, vascular cambium and the roles of each hormone.	Uses most key ideas correctly with only minor errors or missing detail.	Gives few ideas, or uses terms incorrectly, or confuses growth types, hormones or dormancy.
Linking to the Kisumu phenomenon and evidence	Clearly connects every answer to the Kisumu farmer's garden and to class evidence such as the germination containers, mango trunk rings and the germination experiment cups.	Connects most answers to the phenomenon or class evidence.	Gives general answers with little or no link to the phenomenon or evidence.
Explaining cause and effect	Explains how and why, step by step, for example how auxin redistribution causes bending, how increased elongation causes etiolation, and how cambium makes rings.	Gives correct causes but some steps of the reasoning are missing.	States facts without explaining why or how, or the reasoning is incorrect.
Applying ideas to a new problem (Nakuru and regional cases)	Correctly diagnoses etiolation from low light, gives solutions that address the cause, and integrates environment with hormones and growth.	Diagnoses the problem and gives a sensible solution with partial explanation.	Does not identify the cause or gives solutions without scientific reasons.

Clarity and organisation	Answers are complete, well organised and use accurate labelled diagrams or sentences where asked.	Answers are mostly clear and organised, with small gaps.	Answers are unclear, incomplete or hard to follow.
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